Lecture 3 - Generalisation, complexity and error estimation

COMP3670/COMP6670 - Introduction to Machine Learning

Diego Marcondes

Semester 2 2026



Australian National University

Recall the end of Lecture 2

- We built the **statistical model** of supervised learning: a fixed but unknown distribution P generating $(\mathbf{X}, Y)$, a loss ℓ, and the two errors

$$E_{\text{out}}(h) = \mathbb{E}\big[\ell\big((\mathbf{X}, Y), h\big)\big] \qquad E_{\text{in}}(h) = \frac{1}{N}\sum_{i=1}^{N} \ell\big((\mathbf{X}_i, Y_i), h\big)$$

- The learning algorithm is **Empirical Risk Minimisation**: $\hat{h}_N = \arg\min_{h\in\mathcal{H}} E_{\text{in}}(h)$
- And we identified exactly what must be controlled, the **generalisation gap**, uniformly over $\mathcal{H}$:

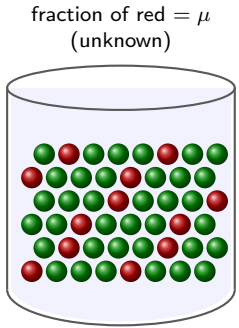
$$\mathbb{P}\Big(\max_{h\in\mathcal{H}} \big|E_{\text{out}}(h) - E_{\text{in}}(h)\big| > \epsilon\Big) \;\leq\; \delta$$

The last missing piece

Proving that such control is **achievable**

Controlling the generalisation gap

A thought experiment: a bin of marbles



- A bin contains red and green marbles in unknown proportions; a fraction μ of them is red
- We draw N marbles **independently** and observe the fraction ν of red ones in our sample
- ν is **known**, μ is **unknown**. Does ν tell us anything about μ?
- Strictly, no: the sample could be all green even if $\mu = 0.99$
- But that is **very unlikely**, and probability lets us say *how* unlikely

- This *is* our problem: fix a hypothesis h, colour an example **red** if h gets it wrong. Then $\mu = E_{\text{out}}(h)$ and $\nu = E_{\text{in}}(h)$

Hoeffding's inequality

Theorem (Hoeffding, 1963)

Let $W_1, \ldots, W_N$ be i.i.d. random variables taking values in $[0, 1]$, with $\mathbb{E}[W_i] = \mu$. Then, for every $\epsilon > 0$,

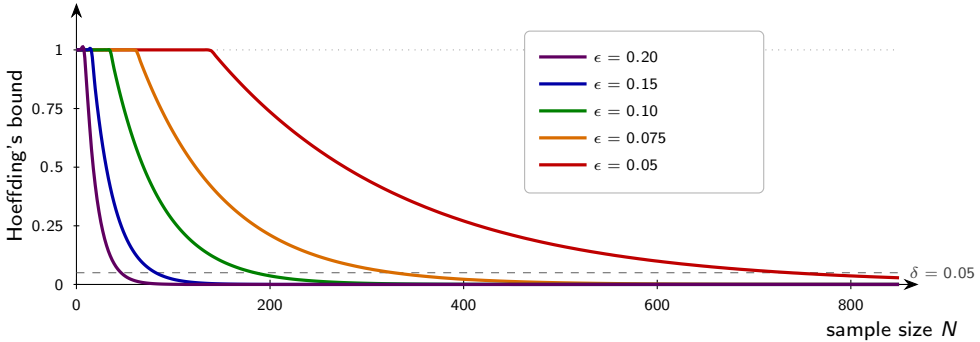
$$\mathbb{P}\left(\left|\frac{1}{N}\sum_{i=1}^{N} W_i \; - \; \mu\right| > \epsilon\right) \; \leq \; 2\, e^{-2N\epsilon^2}.$$

We state it **without proof**: see Devroye, Györfi & Lugosi (1996), *A Probabilistic Theory of Pattern Recognition*, Theorem 8.1.

- Here $e = 2.71828\ldots$ is **Euler's number**, the base of the natural logarithm. The **exponential** e^{-x} is always positive and decreases to 0 *very* fast: $e^{-1} \approx 0.37$, $e^{-5} \approx 0.007$, $e^{-10} \approx 0.00005$
- In words: the **sample average** of bounded i.i.d. quantities is close to its **expectation**, and the probability of being far decays **exponentially** in N
- The power of independence and identical distributions: the bound makes **no assumption** on the distribution of the W_i (it is **distribution-free**), and holds at **every finite** N

Hoeffding's inequality, in a picture

- The bound $2e^{-2N\epsilon^2}$ as a function of the sample size N, one curve for each tolerance ϵ (a probability can never exceed 1, so the bound is only useful once it drops below 1)



- Halving ϵ costs **four times** the data: the curve for $\epsilon = 0.05$ needs $N \approx 740$ to reach $\delta = 0.05$, the one for $\epsilon = 0.10$ only $N \approx 185$

Hoeffding applied to one hypothesis

- Take **binary classification** with the binary loss, and **fix** a hypothesis h. Put

$$W_i = \mathbb{1}\{h(\mathbf{X}_i) \neq Y_i\} \ \in \ \{0,1\} \ \subset \ [0,1]$$

- The W_i are i.i.d. (because the sample is), and

$$\mathbb{E}[W_i] = \mathbb{P}\big(h(\mathbf{X}) \neq Y\big) = E_{\text{out}}(h), \qquad \frac{1}{N}\sum_{i=1}^{N} W_i = E_{\text{in}}(h)$$

Generalisation bound for a single, fixed hypothesis

$$\mathbb{P}\Big(\big|E_{\text{out}}(h) - E_{\text{in}}(h)\big| > \epsilon\Big) \ \leq \ 2\,e^{-2N\epsilon^2}$$

- The bin of marbles, in the language of learning: the bin is the population, red means "h is wrong", $\mu = E_{\text{out}}(h)$ and $\nu = E_{\text{in}}(h)$

Reading the bound

- Set the right-hand side equal to δ and solve for ϵ: $2e^{-2N\epsilon^2} = \delta \iff \epsilon = \sqrt{\frac{1}{2N}\log\frac{2}{\delta}}$
- So, with probability at least $1-\delta$, for any **fixed hypothesis** h

$$E_{\text{out}}(h) \leq \underbrace{E_{\text{in}}(h)}_{\text{computable}} + \underbrace{\sqrt{\frac{1}{2N}\log\frac{2}{\delta}}}_{\text{penalty}}$$

- The penalty decreases like $1/\sqrt{N}$: to **halve** the error bar we need **four times** as much data
- Note this bound that will recur later

Verification is not learning

- Read the previous slide carefully: h had to be **fixed before** looking at the sample
- What we did was **verification** ("this rule I wrote down last year has error at most ..."), **not** learning, in which $\hat{h}$ is *chosen* using the sample
- Why this matters: flip 10 fair coins, 10 times each
  - the probability that a **given** coin shows 10 heads is $2^{-10} \approx 0.001$
  - the probability that **some** coin shows 10 heads is about 0.01, ten times larger
  - with 1 000 coins it is *likely* that some coin shows 10 heads (probability about 0.62), and the record of that coin tells us **nothing** about the fairness of the coin
- ERM does exactly this: it **reports the hypothesis with the best in-sample record**. In the bin analogy, there is now **one bin per hypothesis**, and we deliberately report the bin with the fewest red marbles in the respective sample

The fix for a finite $\mathcal{H}$: the union bound

- Let $\mathcal{H} = \{h_1, \ldots, h_M\}$ be **finite**. The event "*some* hypothesis has a large gap" is a **union** of M events, so the union bound of Lecture 2 applies:

$$\begin{aligned}
\mathbb{P}\left(\max_{h\in\mathcal{H}} \left|E_{\text{out}}(h) - E_{\text{in}}(h)\right| > \epsilon\right) &= \mathbb{P}\left(\bigcup_{j=1}^{M} \left\{\left|E_{\text{out}}(h_j) - E_{\text{in}}(h_j)\right| > \epsilon\right\}\right) \\
&\leq \sum_{j=1}^{M} \mathbb{P}\left(\left|E_{\text{out}}(h_j) - E_{\text{in}}(h_j)\right| > \epsilon\right) \\
&\leq \sum_{j=1}^{M} 2e^{-2N\epsilon^2} = 2M\, e^{-2N\epsilon^2}
\end{aligned}$$

- Each inequality used one fact only: the union bound for the first, Hoeffding for the second

What the finite bound says

- Setting $2Me^{-2N\epsilon^2} = \delta$: with probability at least $1 - \delta$, **simultaneously for every $h \in \mathcal{H}$**,

$$E_{\text{out}}(h) \;\le\; E_{\text{in}}(h) \;+\; \sqrt{\frac{1}{2N}\log\frac{2M}{\delta}}$$

- In particular it holds for $\hat{h}_N$, *whichever* hypothesis the algorithm returned
- The price of searching $\mathcal{H}$ is the term $\log M$: **the complexity of $\mathcal{H}$ enters the bound**
- Sample size needed for $\epsilon = 0.1$ and $\delta = 0.05$ (i.e., 95% confidence):

M (number of hypotheses)	N needed
1	185
100	415
1 000	530

- Encouraging: M grows a *thousandfold* and N less than *triples*, because only $\log M$ matters

But the interesting models are infinite

- The perceptron of Lecture 1 has $\mathcal{H} = \{h_{\mathbf{w}} : \mathbf{w} \in \mathbb{R}^{n+1}\}$, so $M = \infty$ and the bound $2Me^{-2N\epsilon^2}$ is ∞: completely useless
- Yet the perceptron does *not* fail in practice, so the bound must be very **loose**
- Where is it loose? The union bound treats the M events as **unrelated**, adding their probabilities as if they never overlapped
- But two lines that are almost identical make **almost the same mistakes**: if one has a large gap, so does the other. Their events overlap enormously

The key (informal) insight

The sample sees $\mathcal{H}$ **only through the predictions it makes on the N sample points**. Two hypotheses that agree on all N points are **indistinguishable** to the data, and should be counted **once**.

Shattering, shatter coefficient and VC dimension

What the data actually sees

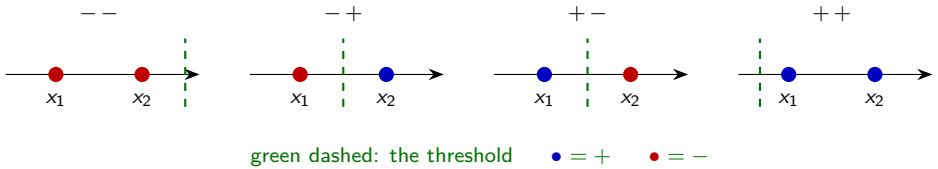
- Consider the case of binary classification, with hypothesis $h : \mathcal{X} \to \{-1, +1\}$
- Fix N points $\mathbf{x}_1, \ldots, \mathbf{x}_N$. Each hypothesis $h \in \mathcal{H}$ produces a **sequence of labels**

$$\big(h(\mathbf{x}_1), h(\mathbf{x}_2), \ldots, h(\mathbf{x}_N)\big) \;\in\; \{+1, -1\}^N$$

- Such a sequence is called a **dichotomy**: it is one way of splitting the N points into two groups
- There are **at most** 2^N dichotomies in total, however large or infinite $\mathcal{H}$ is
- So the "effective number of hypotheses" for a given sample size is not $|\mathcal{H}|$, but the number of **distinct dichotomies** that $\mathcal{H}$ can actually produce
- Let us count them for the simplest model we know: the **perceptron**

The perceptron on a line ($d = 1$): two points

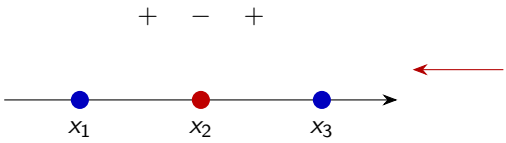
- With one feature, $h(x) = \text{sign}(w_1 x + w_0)$ is a **threshold**: it splits the line at one point, and labels one side $+$ and the other $-$



- All $2^2 = 4$ dichotomies are realisable (the two middle ones use opposite orientations of the threshold)
- We say that the perceptron on a line **shatters** 2 points

The perceptron on a line ($d = 1$): three points

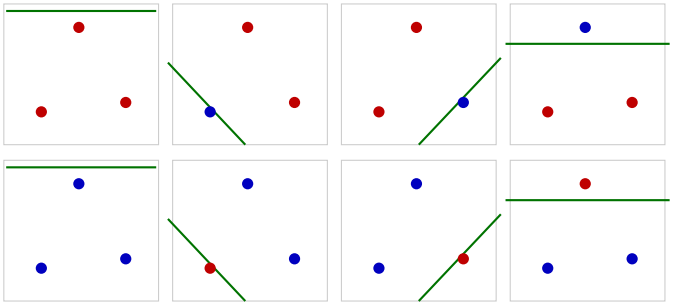
- Now take *any* three points $x_1 < x_2 < x_3$ on the line



- A threshold changes its label **at most once** as we move along the line, so the pattern $+ - +$ (and its mirror $- + -$) can **never** be produced
- So at most $8 - 2 = 6$ of the $2^3 = 8$ dichotomies are realisable: 3 **points are never shattered**
- **Conclusion**: the perceptron on a line shatters 2 points, never 3

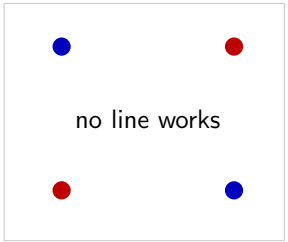
The perceptron in the plane ($d = 2$): three points

- With two features the boundary is a **line**. Take three points not on a common line



- All $2^3 = 8$ dichotomies are produced by a line: the perceptron in the plane **shatters** 3 **points**

The perceptron in the plane ($d = 2$): four points



- This labelling is the **XOR pattern**: the two "diagonal" pairs get opposite labels
- A line cannot produce it, and this is true for **any** four points in the plane, however placed
- So the perceptron in the plane **never shatters 4 points**

	shatters	**never shatters**
Perceptron on a line ($d = 1$)	2 points	3 points
Perceptron in the plane ($d = 2$)	3 points	4 points

The pattern continues: in $\mathbb{R}^d$ the perceptron shatters $d + 1$ points and never $d + 2$.

The shatter coefficient, informally

Shatter coefficient

The **N-th shatter coefficient** $S(\mathcal{H}, N)$ is the **largest number of different dichotomies** that the hypotheses in $\mathcal{H}$ can produce on N points, where we are free to place the N points wherever is most favourable.

- Always $S(\mathcal{H}, N) \leq 2^N$, since there are only 2^N label sequences in total
- When $S(\mathcal{H}, N) = 2^N$ we say that $\mathcal{H}$ **shatters** N points: *some* set of N points can be labelled by $\mathcal{H}$ in *every* conceivable way
- It replaces $M = |\mathcal{H}|$ as the honest count of "how many hypotheses the data can tell apart"
- For the perceptron on a line: $S = 2, 4, 6, 8, \ldots$ for $N = 1, 2, 3, 4, \ldots$ **it stops doubling after $N = 2$**

The Vapnik-Chervonenkis dimension, informally

VC dimension

The **VC dimension** $d_{\mathrm{VC}}(\mathcal{H})$ is the **largest number of points that $\mathcal{H}$ can shatter**. If $\mathcal{H}$ shatters sets of every size, then $d_{\mathrm{VC}}(\mathcal{H}) = \infty$.

- A **single integer** summarising the complexity, or flexibility, of a model
- From the diagrams: $d_{\mathrm{VC}} = 2$ for the perceptron on a line, $d_{\mathrm{VC}} = 3$ in the plane, and $d_{\mathrm{VC}} = d + 1$ in $\mathbb{R}^d$
- Beware:
  - to show $d_{\mathrm{VC}} \geq k$: exhibit **one** set of k points labelled in *all* 2^k ways
  - to show $d_{\mathrm{VC}} < k + 1$: show that **no** set of $k + 1$ points can be labelled in all ways

VC dimensions to know

Hypothesis set (classification) $\mathcal{H}$	$d_{\mathrm{VC}}(\mathcal{H})$	**note**
Thresholds on $\mathbb{R}$, one direction only	1	one parameter
Perceptron on a line ($\mathbb{R}$)	2	1 weight + bias
Intervals $\mathbb{1}\{p_1 \leq x \leq p_2\}$ on $\mathbb{R}$	2	two parameters
Perceptron in $\mathbb{R}^d$	$d+1$	d weights + bias
Neural networks	finite, related to width/depth	very large in practice
All functions $\mathcal{X} \to \{0,1\}$ with $\mathcal{X}$ infinite	∞	unfeasible!

- **Rule of thumb**: d_{VC} is often related to the **number of free parameters**. This is a useful guide, *not* a theorem, and there are counterexamples

Falsifiability, revisited

- Unpack what $d_{\text{VC}}(\mathcal{H}) = \infty$ means: for **every** N, there are N points that $\mathcal{H}$ can label in **all** 2^N possible ways
- Equivalently: **whatever** the sample size, there is some data we may possibly observe such that, **whatever their labels**, $\mathcal{H}$ contains a hypothesis that fits it **perfectly**
- If the unknown distribution P **happens** to concentrate all probability on this kind of **unfalsifiable** data, then the model is **unfalsifiable**. Since we do not know P, we cannot guarantee this is not the case, so we consider this model as **unfalsifiable**

Popper's criterion becomes a theorem

A model with infinite VC dimension is **unfalsifiable**: it is the "secret society" claim of Lecture 1, in mathematical form. And, as we will see, **unfalsifiable** $\iff$ **not learnable**.

- Conversely, $d_{\text{VC}}(\mathcal{H}) = v < \infty$ means that there are datasets with $v + 1$ points that $\mathcal{H}$ **cannot** fit: the model **can** be refuted by data. **That is what makes a good fit meaningful**

The bounds of VC theory and PAC learning

Why one integer is enough

- The reason a single integer suffices is a remarkable **combinatorial** fact

Theorem (Sauer's Lemma)

If $d_{\mathrm{VC}}(\mathcal{H}) = v < \infty$, then for every $N > v$,

$$S(\mathcal{H}, N) \;\leq\; (N+1)^{v}.$$

Stated without proof: see Devroye, Györfi & Lugosi (1996), *A Probabilistic Theory of Pattern Recognition*, Theorem 13.3 and Corollary 13.1.

- The right-hand side is a **polynomial** in N of degree v. So either $\mathcal{H}$ shatters sets of every size and $S(\mathcal{H}, N) = 2^N$ **doubles** at every new point, or $d_{\mathrm{VC}}(\mathcal{H}) = v < \infty$ and $S(\mathcal{H}, N)$ grows only polynomially.
- *e.g.*, perceptron in the plane ($v = 3$), $N = 100$: at most $101^3 \approx 10^6$ labellings, not $2^{100} \approx 10^{30}$
- In the generalisation bounds, $S(\mathcal{H}, N)$ replaces the number of hypotheses M: $(N+1)^v \times e^{-cN\epsilon^2} \to 0$, since **exponential decay beats polynomial growth**

The VC generalisation bound

Theorem (Vapnik and Chervonenkis)

For binary classification with the binary loss, for any distribution P, any N and any $\epsilon > 0$,

$$\mathbb{P}\left(\max_{h\in\mathcal{H}} \left|E_{\text{out}}(h) - E_{\text{in}}(h)\right| > \epsilon\right) \;\leq\; 8\, S(\mathcal{H}, N)\, e^{-N\epsilon^2/32},$$

and therefore, by Sauer's Lemma, if $d_{\text{VC}}(\mathcal{H}) = v < \infty$,

$$\mathbb{P}\left(\max_{h\in\mathcal{H}} \left|E_{\text{out}}(h) - E_{\text{in}}(h)\right| > \epsilon\right) \;\leq\; 8\,(N+1)^v\, e^{-N\epsilon^2/32}.$$

Reference: *Devroye, Györfi & Lugosi (1996), A Probabilistic Theory of Pattern Recognition, Theorem 12.5.*

- Compare with the finite case $2Me^{-2N\epsilon^2}$: the number of hypotheses M has been replaced by $S(\mathcal{H}, N)$, and then by the polynomial $(N+1)^v$
- The constants 8 and 32 are an artefact of the proof; the **shape** is what matters

Reading the VC bound

- As before, set the right-hand side equal to δ: with probability at least $1-\delta$, for *every* $h \in \mathcal{H}$ simultaneously,

$$E_{\text{out}}(h) \;\le\; \underbrace{E_{\text{in}}(h)}_{\text{computable}} \;+\; \underbrace{\sqrt{\frac{32}{N}\left[\, v \log(N+1) + \log\frac{8}{\delta} \right]}}_{\text{complexity penalty } \Omega(N,v,\delta)}$$

- Three readings of the same formula:
  - for **fixed** v, the penalty $\to 0$ as $N \to \infty$, essentially like $\sqrt{v \log N / N}$
  - for **fixed** N, the penalty **grows** with v: more flexible models must be trusted less
  - if $v = \infty$, there is **no bound at all**

From the gap to what we actually care about

- Let $h^\star \in \arg\min_{h\in\mathcal{H}} E_{\text{out}}(h)$ be the **best hypothesis in $\mathcal{H}$**: the best we could ever do inside $\mathcal{H}$, even with infinite data. It is unknown, because P is
- The quality of *learning* in $\mathcal{H}$ is how close $\hat{h}_N$ is to $h^\star$, the **estimation error**

$$\Delta_{\text{est}} \;:=\; E_{\text{out}}(\hat{h}_N) - E_{\text{out}}(h^\star) \;\geq\; 0$$

Theorem

For ERM with the binary loss, if $d_{\text{VC}}(\mathcal{H}) = v < \infty$,

$$\mathbb{P}\Big(E_{\text{out}}(\hat{h}_N) - E_{\text{out}}(h^\star) > \epsilon\Big) \;\leq\; 8\,(N+1)^v\, e^{-N\epsilon^2/128} \;\xrightarrow[N\to\infty]{}\; 0.$$

Reference: *Devroye, Györfi & Lugosi (1996), A Probabilistic Theory of Pattern Recognition, Chapter 12.*

- Why it follows: if *every* h has $|E_{\text{out}} - E_{\text{in}}| \leq \epsilon/2$, then $E_{\text{out}}(\hat{h}_N) \leq E_{\text{in}}(\hat{h}_N) + \frac{\epsilon}{2} \leq E_{\text{in}}(h^\star) + \frac{\epsilon}{2} \leq E_{\text{out}}(h^\star) + \epsilon$, using that $\hat{h}_N$ minimises E_{in} in the middle step

Finite VC dimension is necessary *and* sufficient

The fundamental theorem of Statistical Learning Theory (binary loss)

ERM on $\mathcal{H}$ is **consistent for every distribution** $P \iff d_{\text{VC}}(\mathcal{H}) < \infty$.

"Consistent" means: $E_{\text{out}}(\hat{h}_N) \to E_{\text{out}}(h^\star)$ as $N \to \infty$, with probability one over all possible samples.

Reference: Shalev-Shwartz, S. & Ben-David, S. (2014). *Understanding Machine Learning: From Theory to Algorithms*, Theorem 6.7 (*The Fundamental Theorem of Statistical Learning*); see also Devroye, Györfi & Lugosi (1996), Chapters 12 and 14.

- **Sufficient**: the previous slide
- **Necessary**: if $d_{\text{VC}}(\mathcal{H}) = \infty$, one can *construct* a distribution P, concentrated on points that $\mathcal{H}$ shatters, for which the in-sample error tells us nothing at all
- Combined with the previous section: **learnable** $\iff$ **finite VC dimension** $\iff$ **falsifiable**
- Popper's demarcation criterion for learning from data, in **probabilistic** terms

Turning the bound around: how much data do we need?

- So far we fixed N and asked how large the gap could be. Let us ask the reverse question, the one a practitioner actually asks:

  given a tolerance ϵ and a confidence $1-\delta$, **how many examples do I need?**

- Solving $8(N+1)^{\nu} e^{-N\epsilon^2/32} \leq \delta$ for N gives a number $N_{\mathcal{H}}(\epsilon, \delta)$ that depends on ϵ, δ **and** $d_{\mathrm{VC}}(\mathcal{H})$ **only**
- Crucially, it does **not** depend on the distribution P: the same N works for *every* phenomenon
- This function has a name, the **sample complexity** of $\mathcal{H}$, and the property it expresses has a name too: **PAC learnability**

PAC learning

PAC learnability (Valiant, 1984)

A hypothesis set $\mathcal{H}$ is **PAC learnable** if there exist a learning algorithm and a function $N_{\mathcal{H}}(\epsilon, \delta)$ such that, for every $\epsilon, \delta \in (0, 1)$ and **every** distribution P, if the sample size satisfies $N \geq N_{\mathcal{H}}(\epsilon, \delta)$, then the learned hypothesis $\hat{h}_N$ satisfies

$$\mathbb{P}\Big(E_{\text{out}}(\hat{h}_N) \;\leq\; E_{\text{out}}(h^\star) + \epsilon \Big) \;\geq\; 1 - \delta.$$

- The name is the statement: **P**robably (with probability $\geq 1 - \delta$) **A**pproximately (within ϵ) **C**orrect (as good as the best hypothesis in $\mathcal{H}$)
- What is *not* claimed: not that $\hat{h}_N$ is close to $f^\star$, only that it is close to the best $\mathcal{H}$ has to offer. This is called **agnostic** PAC learning: nothing is assumed about whether $f^\star \in \mathcal{H}$
- Note the **order of the quantifiers**: $N_{\mathcal{H}}(\epsilon, \delta)$ is chosen *before*, and must work for *all* P

Reference: Shalev-Shwartz & Ben-David (2014), Chapter 3.

PAC learning in view of the theory above

- Everything we proved can now be said in one line:

The fundamental theorem, restated

$\mathcal{H}$ is **PAC learnable by ERM** $\iff$ $d_{\mathrm{VC}}(\mathcal{H}) < \infty$ $\iff$ $\mathcal{H}$ is **falsifiable**.

- The sample complexity is obtained from the VC bound, and it grows like

$$N_{\mathcal{H}}(\epsilon, \delta) \;\sim\; \frac{1}{\epsilon^2}\left(v \log \frac{1}{\epsilon} + \log \frac{1}{\delta}\right)$$

- Read off the three dependencies:
  - **linear in the complexity** v: doubling the VC dimension roughly doubles the data needed
  - **quadratic in the accuracy**: halving ϵ costs four times the data
  - **logarithmic in the confidence**: going from 95% to 99.9% costs very little
- Finally, PAC is the **precise form of the probabilistic answer to Hume's problem of induction**: we can never be *certain*, but we can be *probably approximately* right, and we can say how much data that costs at most

Shortcoming 1: the bound is for the *worst* distribution

- The VC bound holds **for every** P, with no assumption whatsoever. That is its great strength, and also its great weakness
- Being valid for *every* distribution means it must be tight for the **worst** one: the most adversarial way the world could be arranged
- Real data is rarely adversarial. Features are typically smooth, redundant and structured, and the actual gap is far smaller than the bound
- Any bound that used properties of the *particular* P at hand would be sharper, but P is unknown, so we cannot compute such a bound

Distribution-free is what makes the bound usable, but it also makes it pessimistic and hence useless.

Shortcoming 2: the sample sizes are unrealistic

- Ask the bound how much data a **perceptron in the plane** ($d_{\mathrm{VC}} = 3$) needs to guarantee a gap of at most $\epsilon = 0.1$ with 95% confidence:

Model	d_{VC}	N **required by the bound**
Perceptron in $\mathbb{R}^2$	3	$\approx 130\,000$
Perceptron in $\mathbb{R}^9$	10	$\approx 430\,000$

- In practice a two-dimensional perceptron generalises well with a **few hundred** points
- So the bound is off by two or three **orders of magnitude**: it is not a design tool for choosing N
- What survives is the **qualitative** content: which quantities matter (N and d_{VC}), and how they trade off against each other

Shortcoming 3: regression is not covered in general

- Everything above used the **binary loss**, which takes values in $\{-1, +1\}$: Hoeffding needs **bounded** random variables
- The theory extends unchanged to any loss bounded by a constant C
- But the **squared loss** $(y - h(\mathbf{x}))^2$ of regression is **unbounded**. If P has heavy tails, a single observation can dominate the average, and *no* distribution-free bound of this type exists
- Bounds for regression *are* available, but they require extra **assumptions on** P (for instance, that the loss has finite moments of some order), so they are no longer distribution-free

The VC-like concepts still apply to regression, but they are more technical, and won't be treated here

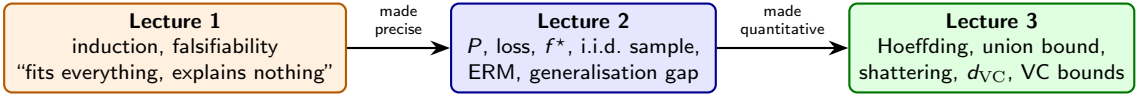
What VC theory *does* give us

The guarantee

If the model is **falsifiable** (finite VC dimension), then with **enough data** the hypothesis returned by ERM is, with high probability, **as good as the best hypothesis in $\mathcal{H}$**. Induction is not made certain, but it is made **quantifiable**.

- It is a **qualitative** theory, and that is not a small thing: it tells us *what* controls generalisation, and it does so before any data is collected
- And it identifies the one design decision that matters most: **the choice of $\mathcal{H}$**

Closing the Statistical Learning Theory overview



"A model so flexible that it can fit anything is unfalsifiable"

$$\Downarrow$$

$$d_{\mathrm{VC}}(\mathcal{H}) < \infty \iff \textbf{the generalisation gap can be bounded} \iff \textbf{PAC-learnable}$$

This theory gives a framework to address the problem of induction from a probabilistic perspective.
And it is all about the **hypothesis set** $\mathcal{H}$.

Bias, variance and the role of the model $\mathcal{H}$

VC theory answers a question we did not quite ask

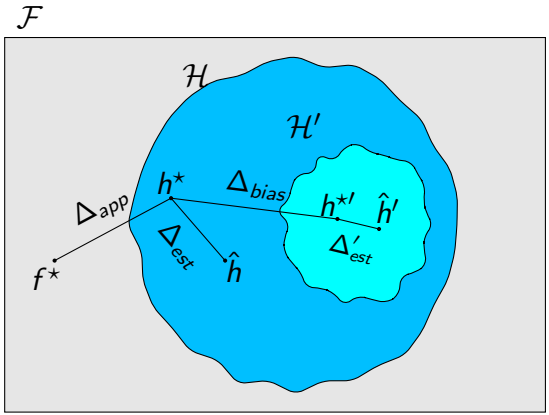
- VC theory guarantees that $E_{\text{out}}(\hat{h}_N)$ approaches $E_{\text{out}}(h^\star)$: **we get the best hypothesis in $\mathcal{H}$**
- But what we *want* is the target function $f^\star$ of Lecture 2, the best rule of all
- If $\mathcal{H}$ is a poor choice, $h^\star$ may be dreadful, and learning it perfectly is of no help
- Write the total error with respect to the best possible, adding and subtracting $E_{\text{out}}(h^\star)$:

The two sources of error

$$\underbrace{E_{\text{out}}(\hat{h}_N) - E_{\text{out}}(f^\star)}_{\text{total error}} = \underbrace{E_{\text{out}}(h^\star) - E_{\text{out}}(f^\star)}_{\Delta_{\text{app}}\text{: approximation error}} + \underbrace{E_{\text{out}}(\hat{h}_N) - E_{\text{out}}(h^\star)}_{\Delta_{\text{est}}\text{: estimation error}}$$

- Δ_{app} is the price of **restricting** to $\mathcal{H}$; Δ_{est} is the price of having only **finite data**

The picture to remember



$\mathcal{F}$: all hypotheses (functions)

$\mathcal{H}, \mathcal{H}'$: two candidate hypothesis sets, with *complexity proportional to size*

$h^\star, h^{\star\prime}$: the best hypothesis in each

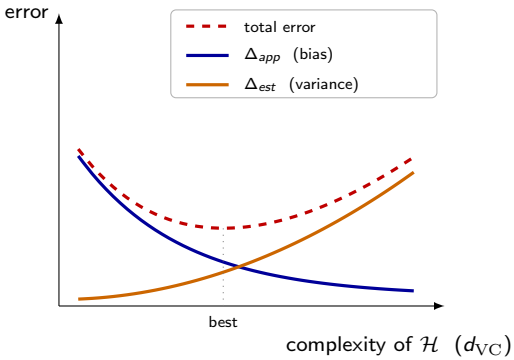
$\hat{h}, \hat{h}'$: the hypotheses actually learned

$f^\star$: the best possible rule (Lecture 2)

Smaller set $\Rightarrow$ smaller Δ'_{est} (easier to learn),
but an extra bias Δ_{bias} to pay

Reference: Marcondes, D.; Simonis, A.; Barrera, J. The role of prior information and computational power in Machine Learning. Figure 1.

The trade-off

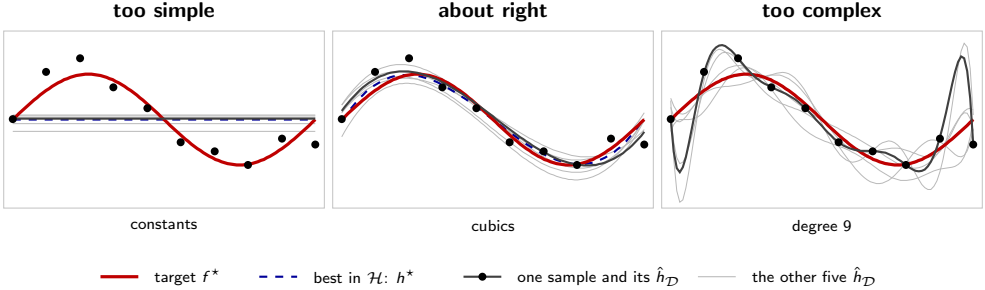


- **Too simple**: small d_{VC}, so little data suffices to learn $h^\star$ well, but $h^\star$ itself is far from $f^\star$ **(underfitting)**
- **Too complex**: $h^\star$ is close to $f^\star$, but with a finite sample we cannot find it **(overfitting)**
- The total error is minimised **somewhere in between**, and where depends on N

More data moves the sweet spot to the right: with more data we can **afford** *a more complex model.*

Bias and variance, in a picture

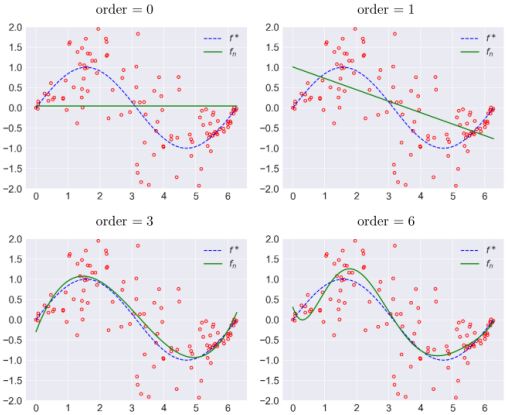
- Target $f^\star(x) = \sin(2\pi x)$, observed at $N = 10$ inputs with noise. **Six independent samples** are fitted in three hypothesis sets. The **dots** are the data of *one* of the six samples, and the **black curve** is the hypothesis learned from it

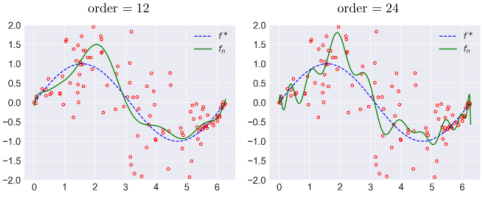


- **Left**: the fits lie on top of each other (tiny variance) but nowhere near red (huge bias)
- **Right**: blue lies *almost exactly* on red (no bias) and the black curve hits *every* dot, yet **no single fit** resembles the target (huge variance)

An empirical example

Polynomials of increasing **order** fitted to *one* noisy sample from $f^\star(x) = \sin(x)$; here f_n is our $\hat{h}_{\mathcal{D}}$





Source: Braga-Neto, U. M. (2024). *Fundamentals of Pattern Recognition and Machine Learning*. 2nd edition. Springer.

Figure 11.2.

The sweet spot

The sweet spot

We want a **simple** hypothesis set that happens to be **close to the target function**.

- Simplicity alone is not enough: a small $\mathcal{H}$ in the wrong place has small variance and enormous bias
- Flexibility alone is not enough either: a huge $\mathcal{H}$ contains something close to $f^\star$, but the data cannot find it
- One way to get **both** is to use what we *know* about the phenomenon to place $\mathcal{H}$ well
- This requires **prior information** and inserting it into the model

Prior information

Back to the example of Lecture 1

- $\mathcal{X} = \{0,1\}^3$: three binary inputs, so $2^3 = 8$ possible inputs in total
- We are given $\mathcal{D}$ with **five** labelled examples, leaving **three** inputs unseen
- $\mathcal{H} =$ **all** functions $\mathcal{X} \to \{0,1\}$, which is $2^8 = 256$ hypotheses
- Lecture 1's conclusion: whatever $\hat{h}$ predicts on those three unseen inputs, exactly 4 of the 8 candidate targets agree on each point and 4 disagree. **The data says nothing**
- In today's language: on $\{0,1\}^3$ this $\mathcal{H}$ shatters all 8 points, so it is as flexible as a model can be, and **it cannot be falsified** by any dataset

x_1	x_2	x_3	y
0	0	0	○
0	0	1	●
0	1	0	●
0	1	1	○
1	0	0	●
1	0	1	?
1	1	0	?
1	1	1	?

Now add one piece of prior information

- Suppose a domain expert tells us:

  "whenever the first variable equals 1*, the outcome is the* **same***, regardless of the other two"*

- This is a statement about the **phenomenon**, made *before* seeing data. It restricts $\mathcal{H}$ to

$$\mathcal{H}' = \{h : h(1, x_2, x_3) \text{ is constant in } (x_2, x_3)\}$$

- The sample contains $(1, 0, 0) \mapsto \bullet$. Combined with the prior information, this **forces** the three unseen labels
- **The problem is solved**: f is now completely determined on all of $\mathcal{X}$

x_1	x_2	x_3	y
0	0	0	○
0	0	1	●
0	1	0	●
0	1	1	○
1	0	0	●
1	0	1	●
1	1	0	●
1	1	1	●

green: the block tied together by the prior information

What the prior information did

- It cut the hypothesis set down:

$$|\mathcal{H}| = 2^8 = 256 \qquad \longrightarrow \qquad |\mathcal{H}'| = 2^4 \times 2 = 32$$

- In terms of complexity, $d_{\mathrm{VC}}(\mathcal{H}) = 8$ became $d_{\mathrm{VC}}(\mathcal{H}') = 5$: the four inputs with $x_1 = 1$ are now **one single block**, which can no longer be labelled freely
- And $\mathcal{H}'$ **is falsifiable**: a dataset containing $(1,0,0) \mapsto \bullet$ and $(1,1,1) \mapsto \circ$ *refutes* it. That is precisely why the fit is now informative
- Note what did **not** happen: we did not collect more data, and we did not change the algorithm

The role of prior information

Prior information is what turns an unlearnable problem into a learnable one. It buys **low bias at low complexity**, which is the only combination that generalises from small samples.

But prior information is usually not that strong

- The example was designed to be extreme: one exact statement, and the problem collapsed
- Real prior information is **vaguer**:
  - "*most* of these 50 measurements are probably irrelevant, but I do not know which"
  - "*some* of these inputs should have the same effect, but I do not know which ones"
  - "the relationship is smooth, but I do not know how smooth"
- Such statements do not single out one $\mathcal{H}$. What they do is single out a **family of plausible candidates**

The idea

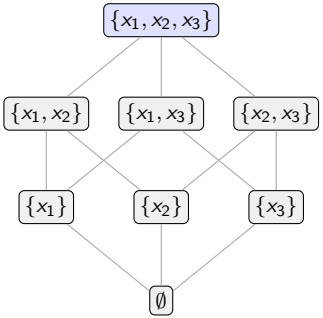
Use the prior information to build a **collection of candidate models** $\mathcal{M}_1, \ldots, \mathcal{M}_n$ inside $\mathcal{H}$, and then let the **data** choose one of them. This is **model selection**.

Model selection

A collection of candidate models

- A **candidate model** is a subset $\mathcal{M} \subset \mathcal{H}$: a smaller, simpler hypothesis set than $\mathcal{H}$ itself
- The first step in model selection is to fix a collection of candidate models $\mathcal{M}_1, \ldots, \mathcal{M}_n$
- Where does the collection come from? **From the prior information**: each $\mathcal{M}_i$ encodes one plausible "story" about the phenomenon
- We want models with **distinct complexities**. In particular, we should have **chains** of models $\mathcal{M}_1 \subset \mathcal{M}_2 \subset \cdots \subset \mathcal{M}_p$ with increasing complexities.

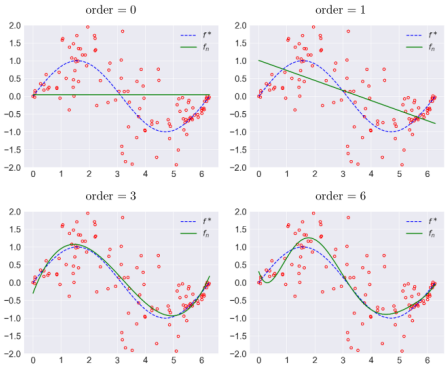
Example 1: variable selection

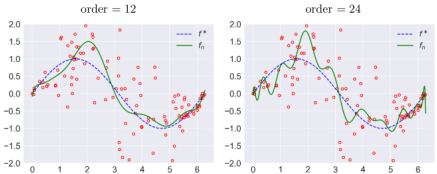


- **Prior information**: only *some* of the input variables matter, but we do not know which
- $\mathcal{M}_A$ = the hypotheses that **depend only on the variables in** A, for $A \subset \{x_1, \ldots, x_d\}$
- The collection is ordered by **inclusion of the variable sets**: $A \subset B \Rightarrow \mathcal{M}_A \subset \mathcal{M}_B$
- Complexity increases upwards: more variables, larger complexity
- There are 2^d candidate models: 8 for $d = 3$, but over a million for $d = 20$

Example 2: polynomial fitting

- **Prior information**: the relationship is smooth, but we do not know how flexible it must be. Candidate models: polynomials of degree $0, 1, 2, \ldots, k$, a **nested chain** $\mathcal{M}_0 \subset \mathcal{M}_1 \subset \cdots \subset \mathcal{M}_k$

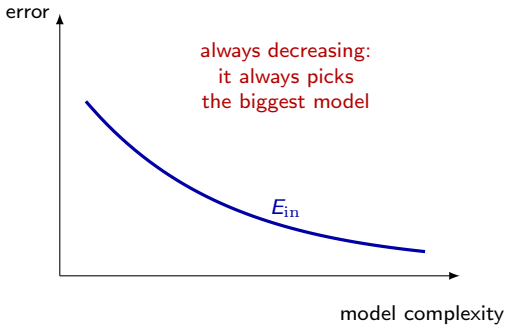




Source: Braga-Neto, U. M. (2024). *Fundamentals of Pattern Recognition and Machine Learning*. 2nd edition. Springer. Figure 11.2.

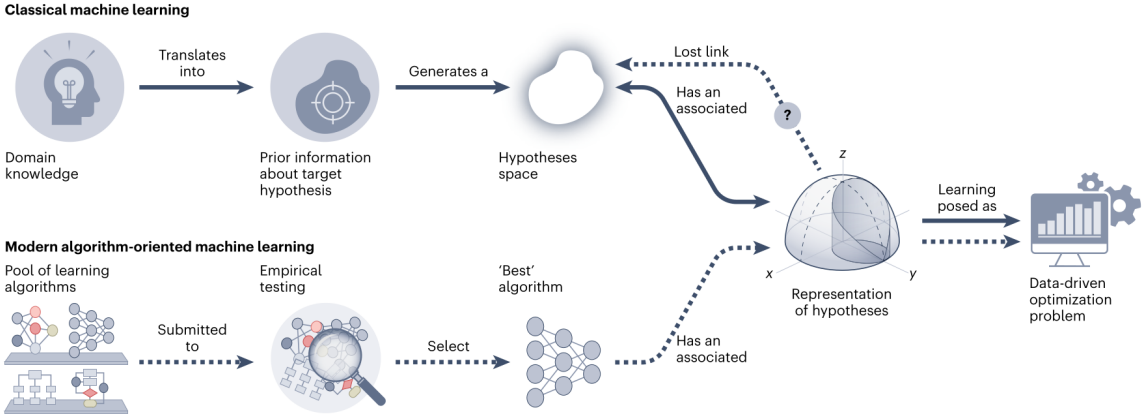
- Orders 0 and 1: high bias. Orders 12 and 24: high variance, the fit moves wildly if one point changes. **Which degree do we pick?**

How *not* to choose: the in-sample error



- If $\mathcal{M}_i \subset \mathcal{M}_j$, then minimising over the larger set can only give a **smaller** in-sample error
- So choosing the model with smallest E_{in} **always** returns the most complex candidate: maximum variance, prone to overfitting
- The in-sample error is **optimistically biased**: the model was chosen using the very data that measures it
- We need an **honest** estimate of E_{out}, computed on data that played *no part* in the fitting

Model selection

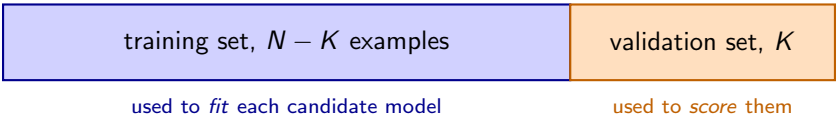


Reference: Marcondes, D.; Simonis, A.; Barrera, J. Back to basics to open the black box. Nature Machine Intelligence. 2024

Validation and cross-validation

The validation set

- Split the dataset in two, **before** any fitting is done:



- Fit $\hat{h}_i$ on the training part, using candidate model $\mathcal{M}_i$, for $i = 1, \ldots, n$
- Score each on the validation part:

$$E_{\text{val}}(\hat{h}_i) \;=\; \frac{1}{K} \sum_{j \in \text{validation}} \ell\big((\mathbf{X}_j, Y_j), \hat{h}_i\big)$$

- Select the model with the smallest validation error

Why the validation error is honest

- The key point: $\hat{h}_i$ was built **without looking at the validation examples**, so relative to them it is a **fixed** hypothesis
- That is exactly the situation of "verification, not learning", where plain Hoeffding applies. With probability at least $1 - \delta$,

$$\left|E_{\text{val}}(\hat{h}_i) - E_{\text{out}}(\hat{h}_i)\right| \;\leq\; \sqrt{\frac{1}{2K}\log\frac{2}{\delta}}$$

- Note what does *not* appear: **no d_{VC}, no complexity term**. The size of $\mathcal{M}_i$ is irrelevant, only K matters
- Concretely, at 95% confidence:

K	200	1 000	5 000
error bar	± 0.096	± 0.043	± 0.019

The dilemma of the split

- Every example put in the validation set is an example **not used for training**
- K **too small**: the estimate E_{val} is unreliable, and we may select the wrong model
- K **too large**: the estimate is precise, but each $\hat{h}_i$ was trained on much less data, so it is worse than the hypothesis we would finally deploy
- Common practical choice: $K \approx N/5$, that is, an 80/20 split
- **Important last step**: once the model $\mathcal{M}_{\hat{i}}$ has been selected, **refit it on the whole dataset** (training + validation). The extra data can only help, and the selection has already been made

One correction: we selected using E_{val}

- We compared n candidates on the validation set and reported the **winner**. That is the coin-flipping problem all over again, on a smaller scale
- The fix is the same: the **union bound** over the n candidates. With probability at least $1-\delta$, for every i,

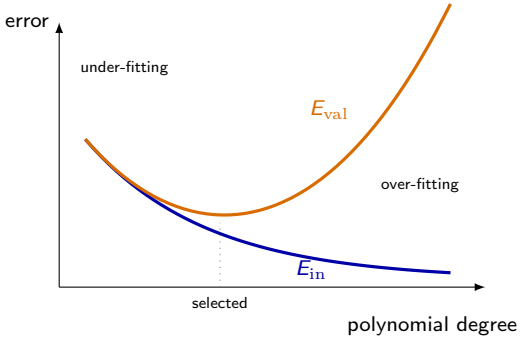
$$E_{\text{out}}(\hat{h}_i) \;\le\; E_{\text{val}}(\hat{h}_i) \;+\; \sqrt{\frac{1}{2K}\log\frac{2n}{\delta}}$$

- The penalty grows only like $\sqrt{\log n}$, so comparing *a few* models is cheap. With $K = 500$ and $\delta = 0.05$:

number of candidates n	1	10	100
error bar	± 0.061	± 0.077	± 0.091

- **Moral**: validation is honest, but only if we do not compare *too many* models on the same validation set

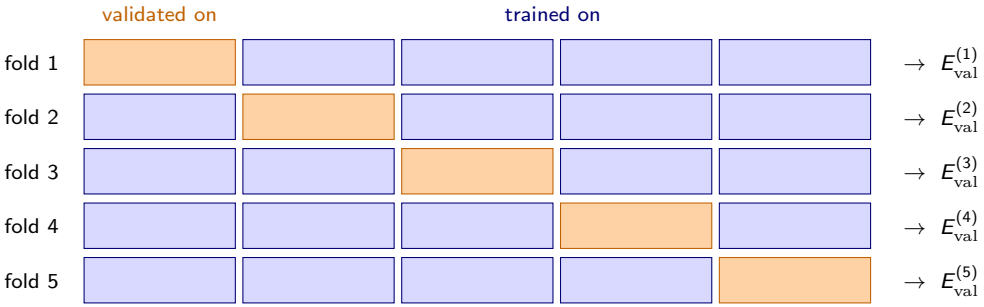
The validation curve



- E_{in} decreases **monotonically**: it is not a model-selection criterion
- E_{val} is **U-shaped**: it first falls (bias decreasing), then rises (variance increasing)
- The minimum of the U is the empirical version of the bias-variance sweet spot
- The **gap** between the two curves is the empirical generalisation gap

Cross-validation: using all the data

- A single split depends on which examples happened to land in the validation set. ***k*-fold cross-validation** rotates the role of the validation part



- Split into k equal folds. For each fold: train on the other $k-1$, score on the held-out one
- Average the k scores: $E_{\text{cv}}(\mathcal{M}_i) = \frac{1}{k}\sum_{j=1}^{k} E_{\text{val}}^{(j)}(\mathcal{M}_i)$, an estimate of E_{out} for model $\mathcal{M}_i$

Cross-validation in practice

- **Every** example is used for training ($k-1$ times) *and* for validation (exactly once)
- Each model is trained on a fraction $\frac{k-1}{k}$ of the data, close to the full sample size when k is not too small
- Averaging k estimates **reduces the variance** of the score, so the comparison between models is more stable
- Common choices: $k = 5$ or $k = 10$. The extreme $k = N$ is **leave-one-out**, but requires N fits
- The cost: k fits per candidate model, so $k \times n$ fits in total
- **Caution**: the k scores are *not* independent (the training sets overlap heavily), so a clean Hoeffding bound is harder to obtain than for a single validation split

The complete model selection procedure

1. Use **prior information** to build the candidate models $\mathcal{M}_1, \ldots, \mathcal{M}_n$
2. For each i, estimate E_{out} for model $\mathcal{M}_i$ by **validation** or **cross-validation**
3. Select $\hat{i} = \arg\min_i E_{\text{cv}}(\mathcal{M}_i)$
4. **Refit** on all the data available for training, using the selected model $\mathcal{M}_{\hat{i}}$, obtaining the final $\hat{h}$

What we have gained

We no longer need to *know* the right hypothesis set. We only need prior information good enough to put a good one **somewhere in the collection**, and then the data picks it out.

And this raises one last question: how good is the $\hat{h}$ we ended up with?

The test sample

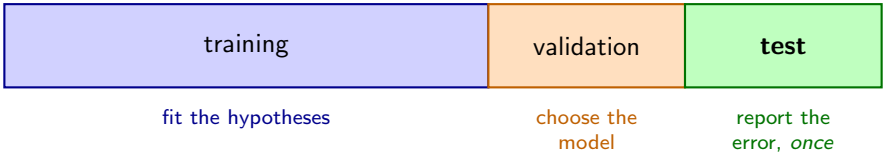
Every error we have computed so far is optimistic

- $E_{\text{in}}(\hat{h})$ is optimistic: $\hat{h}$ was **chosen to minimise it**. This is the whole reason we needed VC theory
- $E_{\text{cv}}(\mathcal{M}_{\hat{i}})$ is *also* optimistic, though much less so: we reported the **smallest** of n scores, and the winner of a competition tends to have been lucky
- Neither can be quoted as "the error of our rule", which is precisely the number a bank, a hospital or a regulator will ask for
- The only cure is data that has taken **no part in any decision whatsoever**: not in fitting, not in selecting, not in tuning

The test sample

A **test sample** is a set of examples, drawn from the same P, that is locked away at the start and looked at **once**, at the very end.

The three-way split



- The final $\hat{h}$ is produced using training + validation only
- Relative to the test sample, $\hat{h}$ is a **single, fixed** hypothesis, so plain Hoeffding applies again: with probability at least $1 - \delta$,

$$\left|E_{\text{test}}(\hat{h}) - E_{\text{out}}(\hat{h})\right| \;\leq\; \sqrt{\frac{1}{2N_{\text{test}}} \log \frac{2}{\delta}}$$

Reading the test error

- **Example**. A classifier is evaluated on $N_{\text{test}} = 1\,000$ held-out examples and misclassifies 83 of them, so $E_{\text{test}}(\hat{h}) = 0.083$
- With $\delta = 0.05$ the error bar is $\sqrt{\frac{1}{2000}\log 40} \approx 0.043$, so with 95% confidence

$$E_{\text{out}}(\hat{h}) \;\in\; [0.040,\; 0.126]$$

- Note that this is **unbiased in both directions**: unlike E_{in}, the test error is as likely to be too pessimistic as too optimistic
- Note also how much sharper it is than the VC bound: no d_{VC} appears, only N_{test}. **Measuring beats bounding** whenever we can afford to hold out data
- And note the cost: those 1 000 examples were **not** used for training

The golden rule

The test set may be used exactly once

The moment you look at the test error and **change something** because of it, the test set has become a validation set, and its estimate is no longer honest.

- "Let me just try one more model and see if the test error improves" is the single most common way of fooling oneself in machine learning
- It is not a matter of discipline but of **mathematics**: Hoeffding needs $\hat{h}$ to be fixed before the test data is seen. Otherwise, the union bound over everything you tried applies
- The same applies at the level of a research community: a public benchmark used by a thousand papers has been implicitly "optimised over" a thousand times
- If you must iterate, iterate on the **validation** set, which exists precisely for that purpose

Where the three errors live

	E_{in}	E_{val} / E_{cv}	E_{test}
computed on	the training data	held-out data	held-out data
used to	*fit* the hypothesis	*choose* the model	*report* the error
biased?	yes, strongly optimistic	mildly optimistic	**no**
error bar involves	d_{VC} and N	n and K (val)	only N_{test}
may be used	always	many times	once

- Three different data sets, three different jobs, three different guarantees
- Confusing them is not a technicality: it is the difference between a rule that works and one that only *appears* to

Lecture highlights

- **Hoeffding's inequality** bounds the generalisation gap of *one fixed* hypothesis; the **union bound** extends it to a *finite* $\mathcal{H}$
- **VC dimension**: the sample sees $\mathcal{H}$ only through the labellings it produces on the N points. The **shatter coefficient** counts them; the **VC dimension** is the largest number of points that can be labelled in *all* ways
- **VC bound**: **PAC learnable** $\iff$ **finite** d_{VC} $\iff$ **falsifiable**. Popper's criterion, as a theorem
- **Two sources of error**: $E_{\mathrm{out}}(\hat{h}) - E_{\mathrm{out}}(f^\star) = \Delta_{\mathrm{app}} + \Delta_{\mathrm{est}}$, the price of restricting $\mathcal{H}$ and the price of finite data
- **Prior information** is what buys low bias *and* low complexity at the same time
- **Validation** and **cross-validation** choose among the candidates; the **test sample** gives the only honest estimate of E_{out}, and must be used **once**

Next lecture: *concrete machine learning models!*

References

- Abu-Mostafa, Y., Magdon-Ismail, M., & Lin, H.-T. (2012). *Learning from Data*. AMLBook.
- Braga-Neto, U. M. (2024). *Fundamentals of Pattern Recognition and Machine Learning*. 2nd edition. Springer.
- Devroye, L., Györfi, L., & Lugosi, G. (1996). *A Probabilistic Theory of Pattern Recognition*. Springer.
- Shalev-Shwartz, S., & Ben-David, S. (2014). *Understanding Machine Learning: From Theory to Algorithms*. Cambridge University Press.
- Valiant, L. G. (1984). A theory of the learnable. *Communications of the ACM*, 27(11), 1134–1142.